

PROBLEM STATEMENT #30 | Smart Cities, Transport & Logistics

Airport Lost Luggage Claim & Tracking Portal

1. Problem Context & Overview

An airline baggage recovery platform that traces bag tag scan timelines across transit hubs, registers lost baggage claims, and manages automated passenger compensation workflows.

Target Stakeholders / Actors: *Passenger, Baggage Service Agent*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.

Sample Acceptance Criteria: Pass: Match identified and recovery alert generated; Fail: Incorrect bag tag format accepted.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.